

Market Microstructure and Electronic Trading Finance 480

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Strategic Trader and Inventory Models

Last lecture we discussed the Roll and sequential trade models:

- Roll model showed volatility and spread may be entangled.
- Sequential trade models assume information asymmetry.
- Glosten-Milgrom model captured some real-world features:
 - Bids, asks, and spreads adjust for adverse selection;
 - Trades have market impact; and,
 - The bid and ask converge to the true V as time increases.
- Easley, Kiefer, O'Hara, and Paperman model added:
 - Trading in continuous time instead of discrete time;
 - Event uncertainty (was V revealed?); and,
 - The probability of informed trading (PIN).
- Hasbrouck casually referred to customer facilitation.

Readings on Today's Topic: More Price Discovery Models

- Chapters 7, 11 of Hasbrouck.
- Chapter 4 of Foucault, Pagano, and Röell.
- Blackboard readings.

For more details: Hasbrouck — on reserve.

I'll repeat two slides from last week.

Remember: We are in a strange world for this lecture.

- Today we'll again slow down, consider financial economics.
- Think about explanations for microstructure phenomena.
- Examine models that explain some features we see in real life.
- This is old material: many papers from 1980s, 1990s.
- This is cutting edge: nascent connections with time series.

For these models, we need a different perspective on the world.

- The models we consider will be highly simplistic.
- Academics rarely explain why we do this — even to students.
- Idea: Simple models let us conduct controlled experiments.
 - Usually: eliminate all but one or two major factors.
 - Question: Does this model reproduce real-life features?
 - If so: the factor we are considering probably matters.
 - We may even have an idea about how that factor matters.
- Simplicity strips away all other excuses for what matters.
- Knowing what matters helps build better models.

All models are wrong; some models are useful.

— George Box

Microstructure Models of Price Mechanics

- Today: finish looking at economic models of price mechanics.
- *Strategic trader* models:
 - Assume asymmetric information (*i.e.* informed traders); and,
 - Traders may trade many times, care about their price impact.
- *Inventory* models consider how dealers (MMs) form prices.

Strategic Trader Models

One-Period Kyle (1985) Model

- Kyle (1985) proposed a model with a single informed trader.
- The informed trader:
 - Considers price impact in setting trade size;
 - Learns security's terminal value v ; and,
 - Submits order for quantity x .
- Liquidity (“noise”) traders submit net order u .
- The single market maker (MM):
 - Observes total order $y = x + u$;
 - Makes up the difference; and,
 - Sets the market clearing price p .
- All trades happen at one price; no bid-ask spread.
- All trading occurs in one period.

One-Period Kyle Model: Informed Trader vs. MM

- The informed trader would like to trade aggressively.
- Q1: What sort of function maps v to order size?
- A: Linear function? (Yes.)
- MM knows larger net orders are more likely to be informed.
- Thus MM sets price increasing in net order size.
- Q2: What sort of function maps order size to MM's price?
- A: Linear function? (Yes.)

One-Period Kyle Model: Setup

$$\text{Security value} \quad v \sim N(p_0, \Sigma_0) \quad (1)$$

$$\text{Noise order} \quad u \sim N(0, \sigma_u^2) \quad u \perp\!\!\!\perp v \quad (2)$$

$$\text{MM assumes: informed order} \quad x = \beta v + \alpha \quad (3)$$

$$\text{Net order} \quad y = x + u \quad (4)$$

$$\text{Informed assumes: trade price} \quad p = \underbrace{\lambda}_{\text{illiquidity}} y + \mu \quad (5)$$

$$\text{Informed trader profit} \quad \pi = (v - p)x \quad (6)$$

$$= (v - \lambda(x + u) - \mu)x \quad (7)$$

One-Period Kyle Model: Optimize, Compute Statistics

If we combine the previous formulæ, we get a little further:

$$E(\pi) = \underbrace{E((v - \lambda(x + u) - \mu)x)}_{E(u)=0; x \text{ non-random}} = (v - \lambda x - \mu)x \quad (8)$$

$$x^* = \operatorname{argmax}_{x \in \mathbb{R}} E(\pi) = \frac{v - \mu}{2\lambda} \quad \text{if } \lambda > 0 \quad (9)$$

$$\Rightarrow \beta = \frac{1}{2\lambda}, \alpha = \frac{-\mu}{2\lambda} \quad (10)$$

$$E(y) = \alpha + \beta E(v) = \frac{-\mu}{2\lambda} + \frac{p_0}{2\lambda} \quad (11)$$

$$\operatorname{Var}(y) = \operatorname{Var}(x) + \operatorname{Var}(u) = \beta^2 \operatorname{Var}(v) + \operatorname{Var}(u) \quad (12)$$

$$= \beta^2 \Sigma_0 + \sigma_u^2 \quad (13)$$

$$\operatorname{Cov}(y, v) = \operatorname{Cov}(\alpha + \beta v, v) = \beta \Sigma_0 \quad (14)$$

One-Period Kyle Model: Find Linear Parameters

Now solve for the linear MM pricing and trader order parameters.

- MM earns no expected profit², prices trade at $p = E(v|y)$.
- Since v, y normal, form of $E(v|y)$ is like linear regression.

$$p = E(v|y) = E(v) + \frac{\text{Cov}(v, y)}{\text{Var}(y)}(y - E(y)) \quad (15)$$

$$= p_0 + \frac{\beta \Sigma_0}{\sigma_u^2 + \beta^2 \Sigma_0}(y - \alpha - \beta p_0) \quad (16)$$

- Use (16), (10), and (5) to solve for $\alpha, \beta, \mu, \lambda$:

$$\alpha = p_0 \frac{\sigma_u}{\sqrt{\Sigma_0}}; \quad \beta = \frac{\sigma_u}{\sqrt{\Sigma_0}}; \quad \mu = p_0; \quad \lambda = \frac{\sqrt{\Sigma_0}}{2\sigma_u}. \quad (17)$$

²Due to competition, again.

One-Period Kyle Model: MM Price; Informed Order, Profit

MM trade price $p = E(v|y) = \lambda y + \mu = \frac{\sqrt{\Sigma_0}}{2\sigma_u} \cdot y + p_0$ (18)

Value uncertainty $\text{Var}(v|y) = \frac{\sigma_u^2 \Sigma_0}{\sigma_u^2 + \beta^2 \Sigma_0} = \frac{\Sigma_0}{2}$ (19)

Informed order $x^* = \beta v + \alpha = \frac{(v - p_0)\sigma_u}{\sqrt{\Sigma_0}}$ (20)

Expected profit $E(\pi) = (v - \lambda x - \mu)x = \frac{(v - p_0)^2 \sigma_u}{2\sqrt{\Sigma_0}}$ (21)

One-Period Kyle Model: Commentary

What can we learn from the one-period Kyle model?

- Trade price linear in net order size, security volatility.
- Trade price inverse to noise order volatility.
- Informed order linear in security's deviation from mean.
- Expected profit quadratic in security's deviation from mean.
 - Large deviations matter much more than small deviations³.
- Informed order, expected profit linear in noise order volatility⁴.
- $\text{Var}(v|y) = \frac{\Sigma_0}{2} \Rightarrow$ Half of information⁵ leaks after one trade.
- Negative net order (*i.e.* $u < -x$) yields negative price. (!)

³Except data errors and adverse selection will hurt you.

⁴Use uninformed orders to hide.

⁵Information in a Fisher sense.

One-Period Kyle Model: Illiquidity Parameter

Finally: Consider the illiquidity parameter.

- Illiquidity parameter $\lambda = \frac{\sqrt{\Sigma_0}}{2\sigma_u}$.
- $\sqrt{\Sigma_0}/\sigma_u$ is ratio of volatilities.
 - Value uncertainty vs. noise order uncertainty.
- $\lambda y = \sqrt{\Sigma_0} \frac{y}{2\sigma_u}$: like liquidity risk:
 - Scaled by volatility of security; and,
 - y/σ_u is similar/proportional to percentage of volume.
- Nice: Demanding liquidity has a cost.
- We will see more such ideas later.

Multi-Period Kyle (1985) Model

- Kyle also discussed a multi-period model.
- Slice time $t \in \{0, 1\}$ into N bins, $n = 1, \dots, N$:

$$\text{Time: } \Delta t_n = 1/N, t_n = n/N. \quad (22)$$

$$\text{Noise order: } \Delta u_n \sim N(0, \sigma_u^2 \Delta t_n). \quad (23)$$

$$\text{Informed order: } \Delta x_n = \beta_n(v - p_{n-1})/N. \quad (24)$$

$$\text{Price change: } \Delta p_n = \lambda_n(\Delta x_n + \Delta u_n). \quad (25)$$

$$\text{E(Later profit): } E(\pi_n | p_{t < n}) = \alpha_{n-1}(v - p_{n-1})^2 + \delta_{n-1} \quad (26)$$

Competition $\Rightarrow p_n = E(v | \Delta x_1 + \Delta u_1, \dots, \Delta x_n + \Delta u_n)$

Informed orders not autocorrelated, by construction.

Multi-Period Kyle Model: Further Definitions

Use the following definitions:

$$\text{Strategy: } X = (x_1, \dots, x_N). \quad (27)$$

$$\text{Pricing rule: } P = (p_1, \dots, p_N). \quad (28)$$

X chosen to always maximize expected future profit:

$$E(\pi_n(X, P) | v, p_1, \dots, p_{n-1}) \geq E(\pi_n(X', P') | v, p_1, \dots, p_{n-1}) \quad n = 1, \dots, N. \quad (29)$$

Multi-Period Kyle Model: Dynamics Equations

Model dynamics are given by difference equations for $n = 1, \dots, N$:

$$\delta_{n-1} = \delta_n + \alpha_n \lambda_n^2 \sigma_u^2 \Delta t_n; \quad (30)$$

$$\alpha_{n-1} = 1/(4\lambda_n(1 - \alpha_n \lambda_n)); \quad (31)$$

$$\beta_n \Delta t_n = \frac{1 - 2\alpha_n \lambda_n}{2\lambda_n(1 - \alpha_n \lambda_n)}; \quad (32)$$

$$\lambda_n = \beta_n \Sigma_n / \sigma_u^2; \quad \text{and,} \quad (33)$$

$$\Sigma_n = (1 - \beta_n \lambda_n \Delta t_n) \Sigma_{n-1}. \quad (34)$$

λ_n is the middle root of the cubic equation:

$$(1 - \lambda_n^2 \sigma_u^2 \Delta t_n / \Sigma_n)(1 - \alpha_n \lambda_n) = \frac{1}{2}. \quad (35)$$

Multi-Period Kyle Model: Solving for Dynamics

Solving for the model dynamics is a bit crusty:

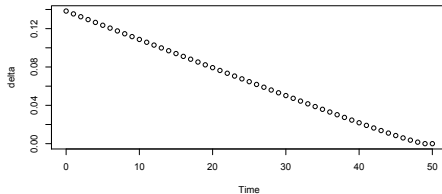
- ① Guess Σ_N (call the guess Σ_N^*).
- ② Use $\alpha_N = \delta_N = 0$ to get $\lambda_N = \frac{\sqrt{\Sigma_N^*}}{\sigma_u \sqrt{2\Delta t_N}}$.
- ③ Set $n = N$.
- ④ Solve for β_n and Σ_{n-1}^* .
- ⑤ Find α_{n-1} for λ_n .
- ⑥ Solve (35)⁶, using middle root for λ_{n-1} .
- ⑦ $n = n - 1$; if $n > 0$, go to step 4.
- ⑧ If $|\Sigma_0^* - \Sigma_0| > \epsilon$: try another Σ_N^* , go to step 1.
- ⑨ Solve for β_0

⁶Numerically or via Cardano's formula.

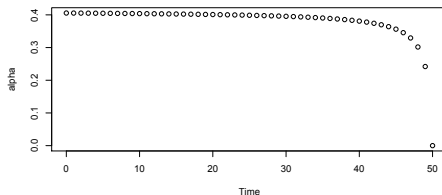
Multi-Period Kyle Model: Simulation

- We can simulate the Kyle model to see its behavior.
- Use $p_0 = 2$, $\Sigma_0 = 0.4$, and $\sigma_u = 0.5$.
- Run one simulation. What do we get?
- The parameter evolution is not so surprising.
- The action evolution is more illuminating.

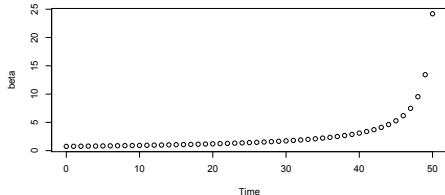
Multi-Period Kyle Model: Evolution of Parameters I



δ : like $E(\text{future profit})$

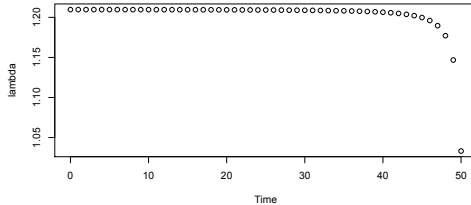


α : fraction of MM mispricing captured

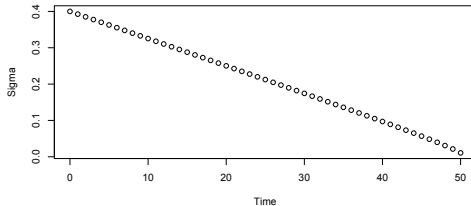


β : aggressiveness of order sizing

Multi-Period Kyle Model: Evolution of Parameters II⁷



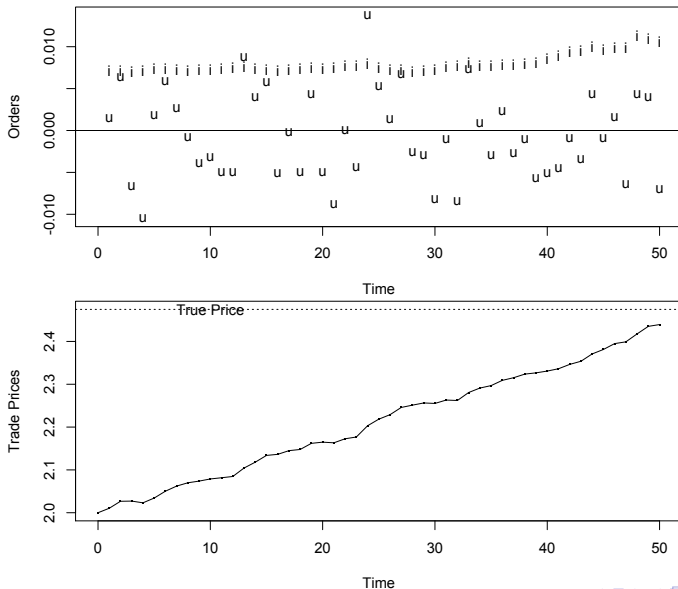
λ : like bid-ask spread



Σ : information not yet revealed

⁷E.B.

Multi-Period Kyle Model: Evolution of Actions



Multi-Period Kyle Model: Commentary

A few details nobody has previously noted⁸:

- Informed order size⁹ increases slightly with time.
- Informed orders are larger after negative uninformed trades.
 - This gets stronger toward end of trade:
 - Compare x_{48} after y_{47} ; y_{30} and x_{31} ; y_4 and x_5 .
- Informed orders decrease after larger net orders, price moves.
 - Recall: one leads to the other; confounding lives here.
- Trade price moves toward the true value.
- Trade price may not converge to true value by end of trading.

⁸As with the Glosten-Milgrom model, I have yet to see plots from anybody else who has simulated the Kyle model.

⁹Informed trades are shown as 'i's; uninformed trades as 'u's.

Admati and Pfleiderer (1988) Model

- Similar to multi-period Kyle model with n_t informed traders.
- Adds a second group of uninformed (discretionary) traders.
- Discretionary traders must trade an exogenous given amount.
- Discretionary traders time¹⁰ trades to reduce cost.
- Security value evolves $\tilde{V} = \bar{V} + \sum_{t=1}^T \tilde{\delta}_t$.
- $\tilde{\delta}_t$ revealed at start of period t .
- At time t , n_t informed traders see $\tilde{\delta}_{t+1} + \epsilon_t$.
- $\tilde{\delta}_t$'s iid, mean 0; forecast variance $\text{Var}(\epsilon_t) = \phi_t$.

¹⁰But they do not slice orders.

Admati-Pfleiderer Model: Results

- Liquidity increases (λ_t decreases) when:
 - More informed or uninformed traders;
 - More forecast or liquidity order uncertainty; or,
 - Lower security value volatility.
- Informed order size increases when:
 - More liquidity order uncertainty; or,
 - Fewer informed traders.
- Discretionaries cluster at highest liquidity order uncertainty¹¹.
- Discretionary clustering attracts informed order clustering¹².
- Constant info revealed/period $\Rightarrow$ constant price variance.
- Discretionaries informative over time: cheaper to trade earlier.

¹¹Discretionaries unite to reduce losses to informeds.

¹²Stronger clustering if informeds can decide to enter market.

- Discretionary traders affect λ_t 's but take them as fixed¹³.
 - Model can be inconsistent.
 - Equilibrium may be fragile.
- If discretionary traders can slice orders:
 - Pure strategy equilibrium may not exist; or,
 - Equilibrium may not be closed-form, unique.
- If informed traders receive differing info:
 - Relationship between n_t and λ_t may be very different.
 - Other effects unclear; trade clustering may disappear.

¹³This is common to many similar models.

Spiegel and Subrahmanyam (1992) Model

- Change: Uninformed adjust orders to price variability.
- Linear MM pricing equilibrium may not exist. (!)
- Equilibrium if uninformed not overwhelmed by informed.
- Liquidity decreases when:
 - Lower risk aversion, net order volatility.
- Liquidity may not be always decrease when:
 - More informed, uninformed traders; forecast volatility.
- Adding liquidity traders may make trading more expensive¹⁴.

¹⁴Similar to trading near an index inclusion.

Inventory Models

Garman (1976) Model

- Single monopolistic market maker.
 - Receives all orders, sets bid and ask;
 - Maximizes expected profit/time s.t. no bankruptcy, failure.
- Failure: MM runs out of cash or securities.
- No market risk; but, infinite horizon.
- MM sets bid p_b and ask p_a once; never changes.
- Buys, sells arrive at Poisson rates $\lambda_a(p), \lambda_b(p)$.
- MM holds $I_c(t)$ units cash, $I_s(t)$ units security.
- $I_c(0) =$ initial cash; $I_s(0) =$ initial securities.

Garman Model: MM Inventory

Let $N_a(t)$ = Cumulative securities sold by MM up to time t ;
 $N_b(t)$ = Cumulative securities bought by MM up to time t .

Then

$$\underbrace{I_c(t)}_{\text{cash inventory}} = \underbrace{I_c(0)}_{\text{initial cash}} + \underbrace{p_a N_a(t)}_{\text{cash from selling}} - \underbrace{p_b N_b(t)}_{\text{cash to buying}} \quad (36)$$

$$\underbrace{I_s(t)}_{\text{security inventory}} = \underbrace{I_s(0)}_{\text{initial security}} + \underbrace{N_b(t)}_{\# \text{ buys}} - \underbrace{N_a(t)}_{\# \text{ sells}}. \quad (37)$$

Garman Model: Failure Probabilities

We also get that:

$$P(\text{Cash failure}) \approx \left(\frac{\lambda_b p_b}{\lambda_a p_a} \right)^{I_c(0)/\bar{p}} \quad \text{if } \lambda_a p_a > \lambda_b p_b \quad (38)$$

$$= 1 \quad \text{else; and,} \quad (39)$$

$$P(\text{Security failure}) \approx \left(\frac{\lambda_a}{\lambda_b} \right)^{I_s(0)} \quad \text{if } \lambda_a < \lambda_b \quad (40)$$

$$= 1 \quad \text{else,} \quad (41)$$

where $\bar{p} = (p_b + p_a)/2$.

Garman Model: Stayin' Alive

- Since certain doom is undesirable, we require that:

$$\lambda_a p_a > \lambda_b p_b \quad \text{and} \quad \lambda_b > \lambda_a. \quad (42)$$

- Thus $p_a > \frac{\lambda_b}{\lambda_a} p_b > p_b$.
- Aha! A bid-ask spread!
- Desire to stay solvent leads to a spread.
- Need not have market risk or adverse selection.
- If MM wants no drift in inventory processes:
 - *i.e.* Wants $E(I_c(t+1)) = E(I_c(t))$ and $E(I_s(t+1)) = E(I_s(t))$.
 - Sets p_b and p_a s.t. $\lambda_a(p_a) = \lambda_b(p_b)$.
 - This policy maximizes profit — but courts disaster¹⁵.

¹⁵Security inventory failure is a recurrent state!

Amihud and Mendelson (1980) Model

- Extension of Garman's (1976) model.
- MM has upper and lower position (*i.e.* credit) limits.
- Eliminates failure concerns.
- Optimal p_b, p_a decrease as $I_c(0), I_s(0)$ increase.
- MM has an optimal inventory position.
- $p_a > p_b$; spread still exists.
- Spread increases with distance from optimal inventory level.
- Spread is not centered about true price of security.
- Profit maximization also leads to a bid-ask spread¹⁶.

¹⁶Economists call this market power — because a profit is made.

Stoll (1978) Model

- MM deviates from target portfolio to trade with others.
- Business costs:
 - Holding suboptimal portfolio;
 - Order processing; and,
 - Adverse selection (trading vs. informed traders).
- Set prices for one buy/sell at time $t = 1$.
- Liquidate trade at time $t = 2$ for true price P_i^* .
- Borrow/lend at risk-free rate R_f .
- Borrowing is unlimited; bankruptcy is not a concern.
- MM has belief about unvarying true price and rate of return.

Stoll Model: Dynamics

Let W_0 = Initial wealth;
 Q_P = Portfolio notional;
 Q_i = True notional of trade in security i ¹⁷;
 $\tilde{R}^*$ = Rate of return on target portfolio;
 $\tilde{R}_i$ = Rate of return on security i ; and,
 C_i = Cost to trade security i .

Then

$$\underbrace{\tilde{W}}_{\text{terminal wealth}} = \underbrace{W_0(1 + \tilde{R}^*)}_{\text{return on target portfolio}} + \underbrace{Q_i(1 + \tilde{R}_i)}_{\text{return for target deviation}} - \underbrace{(Q_i - C_i)(1 + \tilde{R}_i)}_{\text{carry cost for target deviation}}. \quad (43)$$

¹⁷ $Q_i < 0$ if we are selling.

Stoll Model: Trading Condition

- MM only trades if no change in expected utility:

$$E[U(W_0(1 + \tilde{R}^*))] = E[U(\tilde{W})] \quad (44)$$

- Implies relative cost is:

$$\frac{C_i}{Q_i} = c_i = \frac{Z}{W_0}(\rho_{iP}Q_P + \frac{1}{2}\sigma_i^2Q_i). \quad (45)$$

where $U = \text{CARA}^{18}$ utility function;

Z = coefficient of relative risk aversion;

ρ_{iP} = target portfolio–security i correlation; and,

σ_i^2 = variance of security i .

- Similar to stochastic calculus market price of risk.

- Spread for trade notionals $Q_i^a > 0 > Q_i^b$ (MM buy, sell):

$$\frac{P_a - P_b}{P_i^*} = c_i(Q_i^a) - c_i(Q_i^b) = \frac{Z}{W_0} \sigma_i^2 \frac{Q_i^a - Q_i^b}{2} \quad (46)$$

- Higher MM capital decreases spread.
- Higher MM risk aversion increases spread.
- Higher security volatility increases spread.
- Trade size affects transaction costs (higher or lower).
- Spread increases with average trade size.

- Inventory does not affect spread; but:
 - Large (positive) MM inventory decreases bid and ask.
 - Large (negative) MM inventory increases bid and ask.
- Higher correlation to target portfolio increases bid, ask¹⁹.
- Stoll also considered fixed order processing costs:
 - Yields costs which decrease with trade size.
 - Makes MM total cost function U-shaped.

¹⁹ Assuming $Q_P > 0$ — which might not hold.

Ho and Stoll (1981) Model

- Extends Stoll (1978) to multi-period case.
- Time broken into T periods.
- Solve a continuous-time dynamic program for P_b, P_a .
- Trades arrive in continuous-time via Poisson processes.
- Portfolio and base wealth follow drifting random walks²⁰.
- Cash inventory follows a jump process (semimartingale).
- Security inventory follows a jump-diffusion.
 - Oddity: We assume the security price moves around;
 - But, the security has an unvarying true value of P .
 - Ho and Stoll say there is a continuous stock dividend²¹.

²⁰Well, geometric Wiener processes to be picky.

²¹Money market funds do this, but: weird.

Ho-Stoll Model: Results, Issues

Same findings as in Stoll (1978), plus more:

- Spread depends on MM's time horizon to exit trading:
 - Risks decrease as time to exit decreases.
 - Implies cheaper to trade later in the day. (True, slightly.)
 - Even at exit, there is a base risk neutral spread.
- Spread also includes inventory and portfolio risk components.
- Issues:
 - Some approximations used are... not so rigorous.
 - O'Hara finds Poisson arrival rates to be suspect.
 - But the arrival rates change²²; very flexible process.
 - Security price dynamics/constancy could cause trouble.

²²Also known as a *heterogeneous* Poisson process.

- A few interesting themes come out of these models.
- One often overlooked commonality: the need for linearity.
- These models assumed (or examined) linear schemes for:
 - Mapping from alpha to order size; and,
 - Mapping from order flow to MM pricing.
- Huberman and Stanzl (2004): only linear mappings are arbitrage-free.
- Problem: Other evidence indicates mappings are not linear.

The Road Ahead

We're done with focusing on theoretical price models.
Gained intuition about what leads to microstructure phenomena.
Next we'll look statistically at these phenomena.

- Metrics/Models: Prices; Sizes; Limits; Time; Liquidity
- Electronic Trading: Markets; Tools; Tactics; Strategies
- Microstructure and Policy Issues

Readings, in order of importance:

- Blackboard: Readings
- Chapters 12 and 13 of Hasbrouck.
- Chapter 6 of Foucault, Pagano, and Röell.
- Chapters 6 and 8 of O'Hara (not critical)
- Chapter 8 of Johnson (not critical)